

Ammon Brock

ammoncbrock@gmail.com – (208) 789-9477 – www.linkedin.com/in/ammon-brock

Education

MS, Computer Science

May 2026

Brigham Young University

- GPA: 4.00
- Thesis: Methods for Improving Interpretability of Large Language Models (ongoing research)
- Relevant Coursework:

NLP with Transformers	Machine Translation
Machine Learning Operations (AWS)	Advanced Deep Learning
Theory of Predictive Modeling	Advanced Computer Vision

BS, Applied & Computational Mathematics Emphasis (ACME)

April 2024

Brigham Young University

- GPA: 3.99 (Covaledictorian)
- Relevant Coursework:

Machine Learning	Linear Algebra
Advanced Programming Concepts	Multivariable Calculus
Data Structures	Partial Differential Equations
Statistics	Optimal Control
Algorithm Design and Optimization	Time Series Analysis

Skills

- Advanced: Python, C++, Java, Spanish
- Proficient: SQL, Android Programming, HTML
- Specialized Skills:

Natural Language Processing	PyTorch
Pandas	Transformers
Scikit Learn	Active Learning (Human-in-the-loop)
Machine Learning/Deep Learning	Git

Experience

Quantitative Researcher

January 2024 – September 2025

Union Capital Partners – Heber City, UT

- Trained and deployed a Random Forest stock selector used for managing \$100,000 of the fund
- Scraped, downloaded, and stored millions of rows of market data in a MySQL database for efficient retrieval
- Developed a market state classification system used for reducing volatility of trading strategies
- Analyzed hundreds of trading strategies and produced reports quantifying their risk and generalizability

Data Science Intern

May 2023 – October 2023

Zions Bancorporation - Midvale, UT

- Created and deployed a human-in-the-loop command line interface for labeling text data with AI assistance
- Fine-tuned a transformer to classify bank transaction data into 13 categories with over 80% accuracy
- Utilized NLP tools such as spacy and hugging face to reduce unlabeled pool of transaction data by 90%
- Communicated with and implemented suggestions made by users of my data annotation tool

Research Assistant (Data Science)

April 2022 – May 2023

Brigham Young University: Counseling and Psychological Services - Provo, UT

- “Therapist-Client Matching with Machine Learning” -- accepted for publication in Psychotherapy Research
- Used automated machine learning to find a model that improved prediction accuracy by 10%
- Performed data cleaning, model selection, and model validation on data for over 20,000 data entries

Personal Projects

- Trained a multi-modal alignment model on 1.2 million image-caption pairs – outperforms CLIP!
- Deployed an LLM grading assistant website on AWS using Amazon Bedrock, Lambda, S3, CloudFront, etc.
- Built a pipeline for song-to-song English to Spanish translation (audio file to audio file)
- Fine-tuned transformers in multiple domains: bank transactions, twitter sentiment, and stock prediction
- Check out my portfolio! <https://ammonbrock.github.io/portfolio>